NYC 2021 Shootings

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2026-04-07

Introduction

This report looks at NYPD shooting incidents in 2021 and focuses on a few main questions: which boroughs had the most shootings, what time of day shootings were most common, where they tended to happen, and which demographic patterns appeared most often in the available records.

Data Source

The data comes from the NYC Open Data shooting incident dataset: `https://data.cityofnewyork.us/`

This report uses the 2021 portion of the historic NYPD shooting incident data. If the local CSV is not already present, the report downloads it from the NYC Open Data API and stores it in `data/nyc_2021_shootings_2021.csv` before running the analysis.

```
required_packages <- c(
  "dplyr", "ggplot2", "readr", "stringr",
  "tidyr", "tibble", "knitr", "scales"
)
missing_packages <- required_packages[
  !vapply(required_packages, requireNamespace, logical(1), quietly = TRUE)
]

if (length(missing_packages) > 0) {
  install.packages(missing_packages, repos = "https://cloud.r-project.org")
}

invisible(lapply(required_packages, library, character.only = TRUE))
```

```
refresh_data <- FALSE

snapshot_path <- file.path("data", "nyc_2021_shootings_2021.csv")

api_url <- paste0(
  "https://data.cityofnewyork.us/resource/833y-fsy8.csv?",
  "$limit=50000&",
  "$where=occur_date%20between%20%272021-01-01T00:00:00%27%20and%20%272021-12-31T23:59:5
9%27"
)

if (!dir.exists(dirname(snapshot_path))) {
  dir.create(dirname(snapshot_path), recursive = TRUE)
}

if (!file.exists(snapshot_path) || isTRUE(refresh_data)) {
  download.file(api_url, destfile = snapshot_path, mode = "wb", quiet = TRUE)
}

shootings_raw <- readr::read_csv(snapshot_path, show_col_types = FALSE)
```

Data Overview

The dataset contains 2,011 shooting incidents from 2021. It includes the date, time, borough, reported location type, and demographic fields for victims and perpetrators when that information was available.

```
shootings <- shootings_raw %>%
  mutate(
    occur_date = as.Date(occur_date),
    year = as.integer(format(occur_date, "%Y")),
    month = as.integer(format(occur_date, "%m")),
    hour = as.integer(substr(occur_time, 1, 2)),
    boro = stringr::str_to_title(boro),
    location_desc = na_if(location_desc, ""),
    perp_age_group = na_if(perp_age_group, ""),
    perp_sex = na_if(perp_sex, ""),
    perp_race = na_if(perp_race, ""),
    vic_age_group = na_if(vic_age_group, ""),
    vic_sex = na_if(vic_sex, ""),
    vic_race = na_if(vic_race, "")
  )

summary_table <- tibble::tibble(
  metric = c("Rows", "Columns", "Date range"),
  value = c(
    nrow(shootings),
    ncol(shootings),
    paste(min(shootings$occur_date), "to", max(shootings$occur_date))
  )
)

knitr::kable(summary_table)
```

metric	value
Rows	2011
Columns	24
Date range	2021-01-01 to 2021-12-31

```
dplyr::glimpse(shootings)
```

```
## Rows: 2,011
## Columns: 24
## $ incident_key            <dbl> 238487793, 238490103, 238481445, 238481446, 23…
## $ occur_date              <date> 2021-12-31, 2021-12-31, 2021-12-31, 2021-12-3…
## $ occur_time              <time> 15:36:00, 19:23:00, 04:25:00, 04:00:00, 01:05…
## $ boro                    <chr> "Brooklyn", "Staten Island", "Brooklyn", "Bron…
## $ loc_of_occur_desc       <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA…
## $ precinct                <dbl> 75, 120, 70, 40, 25, 32, 112, 49, 48, 43, 115,…
## $ jurisdiction_code       <dbl> 2, 0, 0, 2, 2, 0, 0, 0, 0, 0, 0, 0, 2, 2, 0, 0…
## $ loc_classfctn_desc      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA…
## $ location_desc           <chr> "MULTI DWELL - PUBLIC HOUS", "PVT HOUSE", NA, …
## $ statistical_murder_flag <lgl> FALSE, TRUE, FALSE, FALSE, FALSE, FALSE, FALSE…
## $ perp_age_group          <chr> NA, "45-64", NA, NA, NA, NA, NA, NA, NA, NA, N…
## $ perp_sex                <chr> NA, "M", NA, NA, NA, NA, NA, NA, NA, NA, NA, N…
## $ perp_race               <chr> NA, "WHITE", NA, NA, NA, NA, NA, NA, NA, NA, N…
## $ vic_age_group           <chr> "25-44", "25-44", "25-44", "25-44", "<18", "25…
## $ vic_sex                 <chr> "M", "M", "M", "M", "F", "M", "M", "F", "M", "…
## $ vic_race                <chr> "BLACK", "WHITE", "BLACK", "WHITE HISPANIC", "…
## $ x_coord_cd              <dbl> 1016133, 966127, 995435, 1007753, 1003188, 999…
## $ y_coord_cd              <dbl> 178623, 162589, 174621, 235353, 231190, 238251…
## $ latitude                <dbl> 40.65691, 40.61294, 40.64597, 40.81264, 40.801…
## $ longitude               <dbl> -73.88509, -74.06528, -73.95970, -73.91510, -7…
## $ geocoded_column         <chr> "POINT (-73.88509115599999 40.65690638000007)"…
## $ year                    <int> 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021…
## $ month                   <int> 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12, 12…
## $ hour                    <int> 15, 19, 4, 4, 1, 19, 3, 22, 19, 21, 20, 20, 19…
```

Data Notes

Some fields are missing, especially in the perpetrator columns. Because of that, the report keeps the full dataset and only filters missing values in sections that need those specific fields.

```
missing_summary <- shootings %>%
  summarise(
    across(
      c(
        boro, location_desc,
        perp_age_group, perp_sex, perp_race,
        vic_age_group, vic_sex, vic_race
      ),
      ~ sum(is.na(.x))
    )
  ) %>%
  tidyr::pivot_longer(
    everything(),
    names_to = "field",
    values_to = "missing_values"
  ) %>%
  arrange(desc(missing_values))

knitr::kable(missing_summary)
```

field	missing_values
location_desc	1396
perp_age_group	1049
perp_sex	1049
perp_race	1049
boro	0
vic_age_group	0
vic_sex	0
vic_race	0

Perpetrator and Victim Profiles

The most common pattern in the available records appears among Black men between 25 and 44. That group shows up at the top of both the perpetrator and victim summaries. Other common groups include younger Black men and White Hispanic men in similar age ranges.

```
perpetrators <- shootings %>%
  filter(
    !is.na(perp_race),
    !is.na(perp_sex),
    !is.na(perp_age_group)
  ) %>%
  count(perp_race, perp_sex, perp_age_group, sort = TRUE)

knitr::kable(head(perpetrators, 10))
```

perp_race	perp_sex	perp_age_group	n
BLACK	M	25-44	323
BLACK	M	18-24	202
WHITE HISPANIC	M	25-44	76
WHITE HISPANIC	M	18-24	70
BLACK	M	<18	64
BLACK HISPANIC	M	25-44	58
BLACK HISPANIC	M	18-24	38
BLACK	M	45-64	32
WHITE HISPANIC	M	<18	16
BLACK	F	18-24	12

```
victims <- shootings %>%
  filter(
    !is.na(vic_race),
    !is.na(vic_sex),
    !is.na(vic_age_group)
  ) %>%
  count(vic_race, vic_sex, vic_age_group, sort = TRUE)

knitr::kable(head(victims, 10))
```

vic_race	vic_sex	vic_age_group	n
BLACK	M	25-44	698
BLACK	M	18-24	380
WHITE HISPANIC	M	25-44	144
BLACK	M	<18	115
BLACK HISPANIC	M	25-44	111
BLACK	M	45-64	104

vic_race	vic_sex	vic_age_group	n
WHITE HISPANIC	M	18-24	74
BLACK HISPANIC	M	18-24	66
BLACK	F	25-44	54
BLACK	F	18-24	41

These summaries describe the reported incidents, but they do not explain the full cause of the violence on their own. They make more sense when read together with the borough and location patterns below.

Shootings by Borough

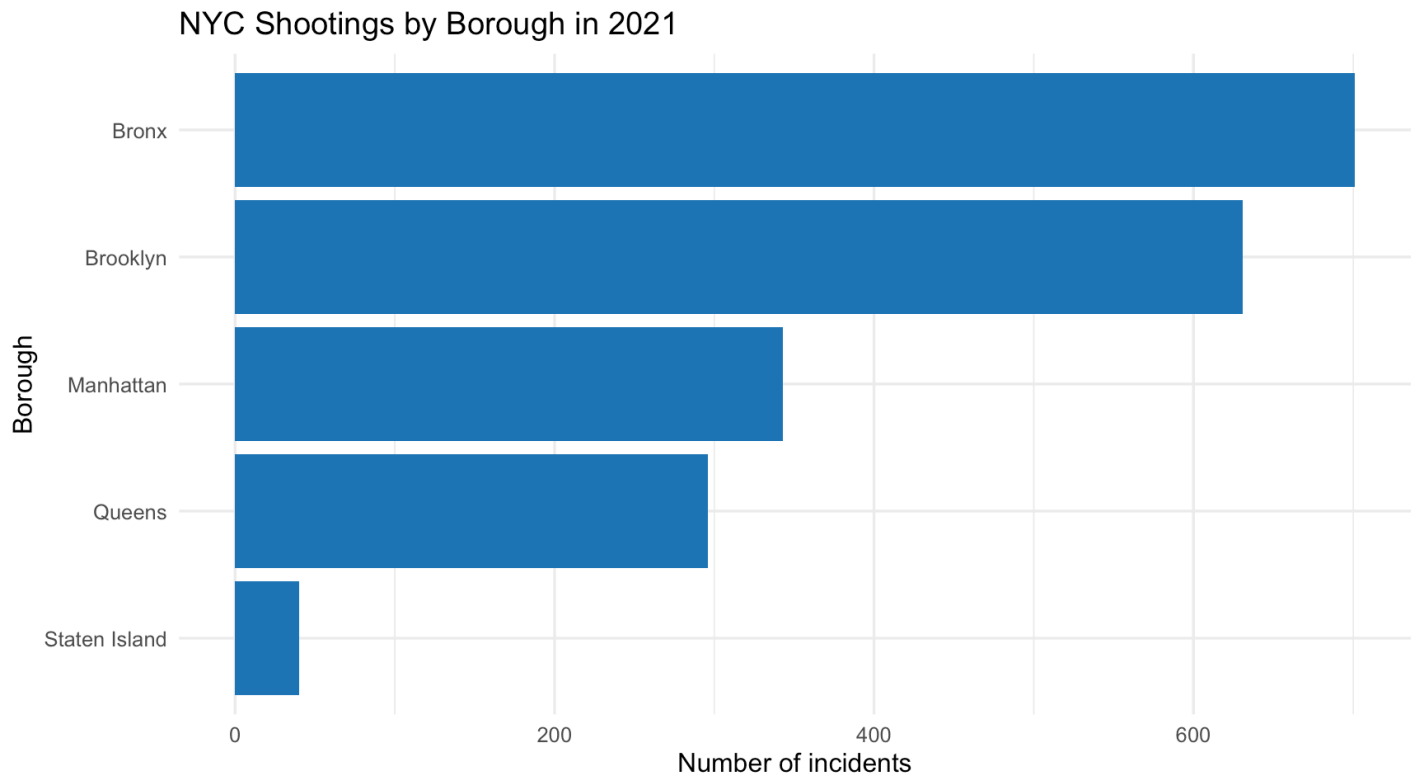
The borough totals show that the Bronx had the most shootings in 2021, followed by Brooklyn. Manhattan and Queens had fewer incidents, and Staten Island had far fewer than the other boroughs.

```
borough_summary <- shootings %>%
  filter(!is.na(boro)) %>%
  count(boro, sort = TRUE)

knitr::kable(borough_summary)
```

boro	n
Bronx	701
Brooklyn	631
Manhattan	343
Queens	296
Staten Island	40

```
ggplot(borough_summary, aes(x = reorder(boro, n), y = n)) +
  geom_col(fill = "#1f78b4") +
  coord_flip() +
  scale_y_continuous(labels = scales::label_comma()) +
  labs(
    title = "NYC Shootings by Borough in 2021",
    x = "Borough",
    y = "Number of incidents"
  ) +
  theme_minimal(base_size = 12)
```



The Bronx stands out clearly in the chart. Shootings were not spread evenly across the city, and some neighborhoods carried much more of the burden than others.

Shootings by Time of Day

The time pattern shows that shootings were most common late at night. The highest counts appear around 11 PM, with high totals continuing through the nearby overnight hours.

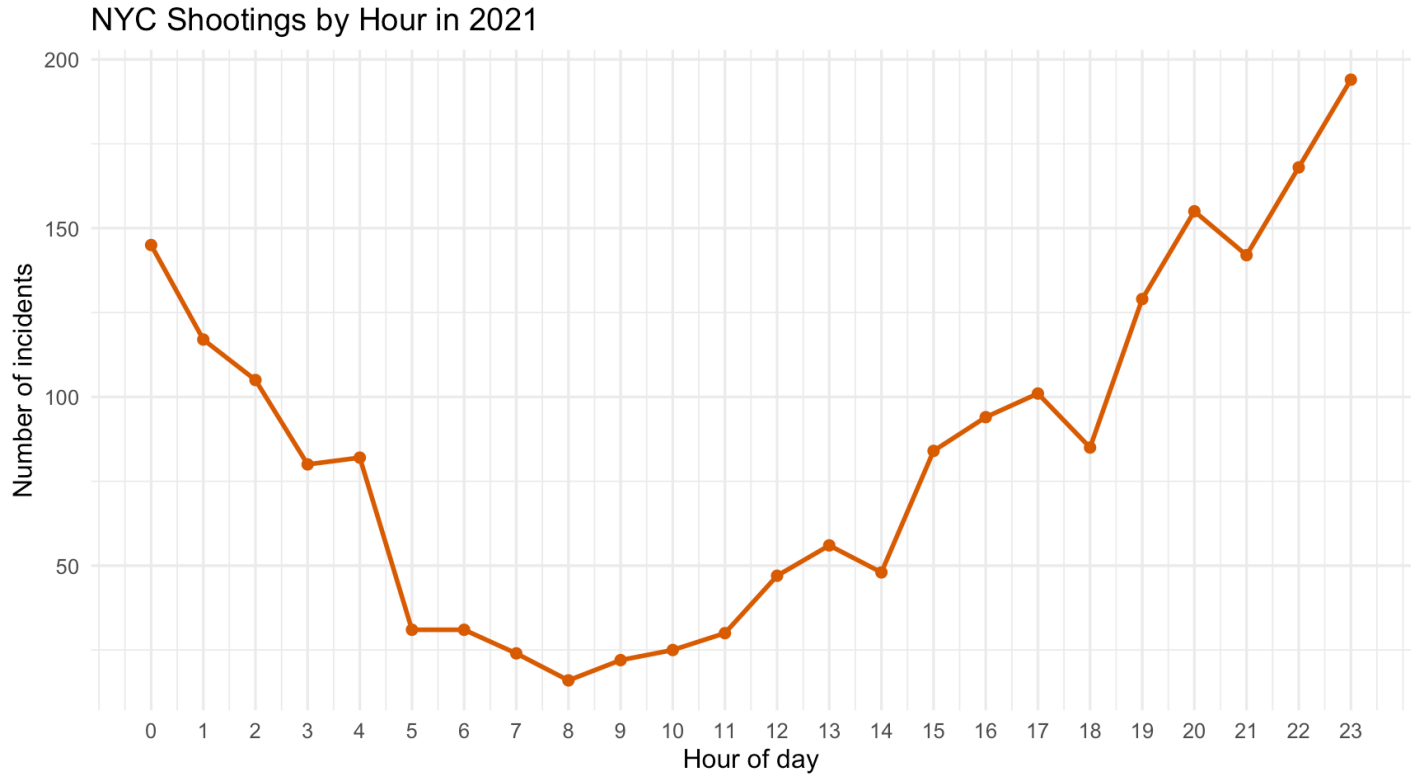
```
hourly_summary <- shootings %>%
  filter(!is.na(hour)) %>%
  count(hour, sort = FALSE)

knitr::kable(hourly_summary)
```

hour	n
0	145
1	117
2	105
3	80
4	82
5	31
6	31
7	24

hour	n
8	16
9	22
10	25
11	30
12	47
13	56
14	48
15	84
16	94
17	101
18	85
19	129
20	155
21	142
22	168
23	194

```
ggplot(hourly_summary, aes(x = hour, y = n)) +
  geom_line(color = "#d95f02", size = 1) +
  geom_point(color = "#d95f02", size = 2) +
  scale_x_continuous(breaks = 0:23) +
  scale_y_continuous(labels = scales::label_comma()) +
  labs(
    title = "NYC Shootings by Hour in 2021",
    x = "Hour of day",
    y = "Number of incidents"
  ) +
  theme_minimal(base_size = 12)
```



This is one of the clearest patterns in the data. Many incidents happened when the city was quieter and visibility was lower.

Shootings by Reported Location

The location descriptions show that many incidents were reported in public housing and apartment buildings. That suggests a large share of shootings happened in residential spaces, not only in open public areas.

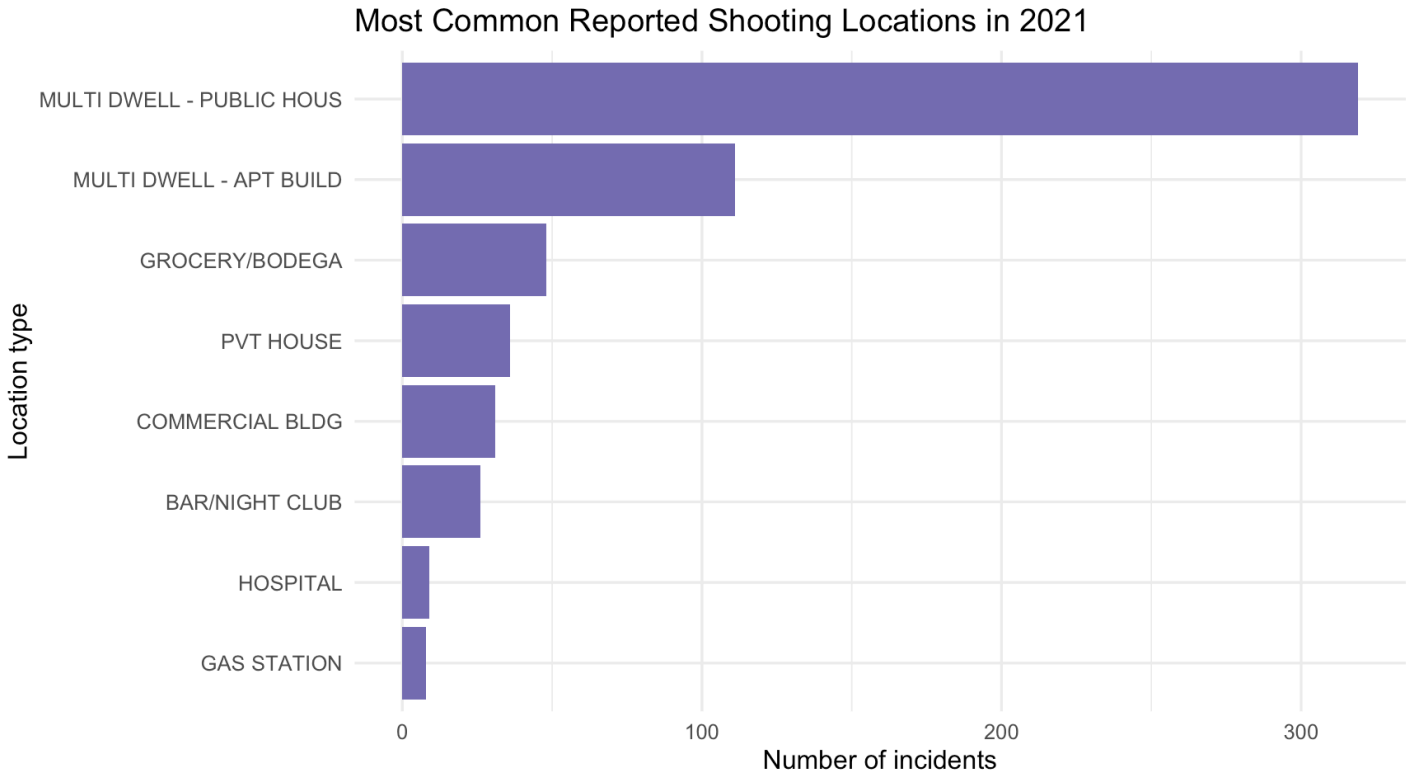
```
location_summary <- shootings %>%
  filter(!is.na(location_desc)) %>%
  count(location_desc, sort = TRUE)

knitr::kable(head(location_summary, 10))
```

location_desc	n
MULTI DWELL - PUBLIC HOUS	319
MULTI DWELL - APT BUILD	111
GROCERY/BODEGA	48
PVT HOUSE	36
COMMERCIAL BLDG	31
BAR/NIGHT CLUB	26
HOSPITAL	9
GAS STATION	8

location_desc	n
HOTEL/MOTEL	8
RESTAURANT/DINER	6

```
ggplot(head(location_summary, 8), aes(x = reorder(location_desc, n), y = n)) +
  geom_col(fill = "#7570b3") +
  coord_flip() +
  scale_y_continuous(labels = scales::label_comma()) +
  labs(
    title = "Most Common Reported Shooting Locations in 2021",
    x = "Location type",
    y = "Number of incidents"
  ) +
  theme_minimal(base_size = 12)
```



Public housing developments and apartment buildings appear most often in the location field. That fits the broader picture of shootings being concentrated in residential parts of the city.

Victim and Perpetrator Combinations

Looking across borough, victim race, perpetrator race, and age groups together helps show which combinations appeared most often in the records with more complete information.

```
incident_combinations <- shootings %>%
  filter(
    !is.na(boro),
    !is.na(vic_race),
    !is.na(vic_age_group),
    !is.na(perp_race),
    !is.na(perp_age_group)
  ) %>%
  count(boro, perp_race, vic_race, perp_age_group, vic_age_group, sort = TRUE)

knitr::kable(head(incident_combinations, 10))
```

boro	perp_race	vic_race	perp_age_group	vic_age_group	n
Bronx	BLACK	BLACK	25-44	25-44	58
Brooklyn	BLACK	BLACK	25-44	25-44	53
Queens	BLACK	BLACK	25-44	25-44	32
Manhattan	BLACK	BLACK	25-44	25-44	28
Brooklyn	BLACK	BLACK	18-24	18-24	24
Bronx	BLACK	BLACK	18-24	25-44	22
Manhattan	BLACK	BLACK	18-24	25-44	21
Brooklyn	BLACK	BLACK	25-44	18-24	20
Bronx	BLACK	BLACK	18-24	18-24	17
Bronx	BLACK	BLACK	25-44	18-24	17

These combined summaries show repeated patterns in the neighborhoods most affected by shootings. They suggest the issue is tied to place and circumstance, not just isolated incidents.

Conclusion

The data shows a few clear patterns. The Bronx had the highest number of incidents, shootings were most common late at night, and many were reported in public housing and apartment buildings. In the available demographic records, Black men between 25 and 44 appeared most often among both perpetrators and victims.

These patterns suggest the problem is closely tied to neighborhood conditions and to the environments in which people live. Poverty, education, housing conditions, and long-term disinvestment are more useful ways to think about the violence than simple demographic labels alone. The value of the data is that it helps track where the violence is concentrated and can support better decisions about prevention and community investment.

Session Notes

```
sessionInfo()
```

```
## R version 4.2.2 (2022-10-31)
## Platform: aarch64-apple-darwin20 (64-bit)
## Running under: macOS 14.6.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/lib/libRblas.0.d
ylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/lib/libRlapack.d
ylib
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] scales_1.2.1  knitr_1.42    tibble_3.1.8  tidyr_1.3.0   stringr_1.5.0
## [6] readr_2.1.4   ggplot2_3.4.1 dplyr_1.1.0
##
## loaded via a namespace (and not attached):
##  [1] highr_0.10       pillar_1.8.1     bslib_0.4.2      compiler_4.2.2
##  [5] jquerylib_0.1.4  tools_4.2.2      bit_4.0.5        digest_0.6.31
##  [9] jsonlite_1.8.4   evaluate_0.20    lifecycle_1.0.3  gtable_0.3.1
## [13] pkgconfig_2.0.3  rlang_1.0.6      cli_3.6.0        rstudioapi_0.14
## [17] parallel_4.2.2   yaml_2.3.7       xfun_0.37        fastmap_1.1.0
## [21] withr_2.5.0      generics_0.1.3   vctrs_0.5.2      sass_0.4.5
## [25] hms_1.1.2        bit64_4.0.5      grid_4.2.2       tidyselect_1.2.0
## [29] glue_1.6.2       R6_2.5.1         fansi_1.0.4      vroom_1.6.1
## [33] rmarkdown_2.20   farver_2.1.1     purrr_1.0.1      tzdb_0.3.0
## [37] magrittr_2.0.3   htmltools_0.5.4  ellipsis_0.3.2   colorspace_2.1-0
## [41] labeling_0.4.2   utf8_1.2.3       stringi_1.7.12   munsell_0.5.0
## [45] cachem_1.0.6     crayon_1.5.2
```